

WESLEY LUU

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DoD Secret (Active) | CompTIA Security+

Education

California Polytechnic State University, San Luis Obispo

June 2024

Bachelor of Science in Computer Engineering, Cum Laude

San Luis Obispo, CA

- Relevant Coursework: Semiconductor Devices, Digital Electronics & ICs, Circuit Analysis, Signals and Systems, Computer Architecture & Design, Parallel Computing, Systems Programming, Operating Systems, Microcontrollers & Embedded Applications

Technical Skills

Languages	C, Python, SystemVerilog, TypeScript, Java, SQL
Software	React, Next.js, .NET, AWS, Linux
Embedded Systems	STM32, RISC-V, Interrupts
Protocols	SPI, I2C, UART, TCP/IP, UDP
Tools	LTSpice, Git, Docker, Vivado, Oscilloscope, Logic Analyzer, Multimeter

Experience

Software Engineer

April 2025 – Present

RGNext | C, Python, SQL, .NET

Vandenberg, CA

- Architected and built a DoD onboarding platform that centralized access requests, approvals, document collection, and user tracking into a single self-service system, reducing administrative workload by **30%**.
- Implemented secure full-stack workflows using .NET, CAC/X.509 auth, and automated approval pipelines across two integrated web applications.
- Modernized a legacy post-launch analysis workflow by building a C-based application that ran Fortran simulation code directly on analysis machines, eliminating manual floppy-disk data transfers.

Software Engineer

June 2024 - March 2025

Synergistic Intelligence | AWS, React, TypeScript

San Luis Obispo, CA

- Built full-stack product features using modular React components and backend APIs, supporting multiple user journeys and scalable feature delivery.
- Deployed and maintained the application on AWS using services such as Elastic Beanstalk, EC2, and S3, improving reliability and streamlining production releases.

Intern Software Engineer

April 2023 – June 2024

California Cybersecurity Institute | Python, React, AWS

San Luis Obispo, CA

- Created from scratch a support ticket system that verified over 1K+ students and handled over 1K+ support tickets for their cybersecurity competition.
- Redesigned the user interface of existing pages, achieving a more intuitive design that halved the volume of support tickets related to those pages by 50%.

Projects

Latch | Python, TypeScript

March 2026 - Present

- Built a TUI tool for Claude Code that leverages IPC and hooks to eliminate context-switching by keeping session history and git state without leaving the terminal
- A TUI with a side-by-side diff viewer, session plan reader, and searchable project picker by parsing message blocks, tool calls, and token metadata to power session search, diff inspection, and workspace navigation.

DemoStudios | TypeScript, React, SQL

January 2024 - Present

- Developed and styled front-end pages for a React-based music-sharing website, closely following Figma designs
- Engineered search filtering functionality using SQL queries, significantly improving search response times by **30%**, which enhanced user engagement and satisfaction by streamlining access to hundreds of songs and stems.

Hack4Impact | Typescript, Next.js, AWS, MongoDB

September 2024 - June 2024

- Increased scheduling efficiency for non-profit org by **30%** by developing a full-stack Calendar app.
- Architected responsive frontend features and implemented backend solutions with MongoDB, creating RESTful APIs for user and event management.

Quanser ACC Autonomous Vehicle Competition

March 2024 - June 2024

- **Achievement:** Advanced to Quanser ACC International Student Competition, earning a spot among the **top 10** teams out of 40
- Designed and implemented a Kalman filter for multi-sensor fusion, improving localization accuracy and vehicle state estimation by about 15% over raw sensor measurements.